Series XVII – Article XVII.3: Pillar P2 – Constant Spectral Gap via Kithima Bridge

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Abstract

We prove that the spectral Hamiltonian $H_{n,k} = \hat{V}_{n,k} + \Delta$ for the maximal determinant problem has a constant spectral gap $\gamma(H_{n,k}) \geq 2$. The diagonal potential $\hat{V}_{n,k}$ is non-negative, and Δ is the adjacency matrix of the hypercube, which has spectral gap $\gamma(\Delta) = 2$. The Kithima Bridge lemma states that adding a non-negative diagonal potential cannot decrease the spectral gap. Therefore $\gamma(H_{n,k}) \geq \gamma(\Delta) = 2$. This constant gap is independent of n and k and guarantees exponential convergence of the power iteration algorithm (PillarP4). All steps are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Articles XVII.1 and XVII.2 constructed the spectral Hamiltonian $H_{n,k} = \hat{V}_{n,k} + \Delta$ and proved that it has a unique strictly positive ground state ψ_0 (P1). In this article we establish a uniform lower bound on the spectral gap.

2 Recap of the Hamiltonian

From Article XVII.2:

- $\Omega = \{0, 1\}^M$, M variables in $\Phi_{n,k}$.
- Diagonal potential $\hat{V}_{n,k}(\sigma) = 0$ for satisfying assignments (matrices with $|\det H| \geq k$), M^2 otherwise.

- Δ: adjacency of the hypercube, with eigenvalues $M - 2k$, $k = 0, \ldots, M$, thus $\gamma(\Delta) = 2$.
- $H_{n,k} = \hat{V}_{n,k} + \Delta$.

3 The Kithima Bridge lemma

Lemma 3.1 (Kithima Bridge)**.** *Let V be a diagonal matrix with non-negative entries, and let Δ be a symmetric matrix with non-negative off-diagonal entries. Then*

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Proof. The proof uses the Courant-Fischer minimax theorem. For any vector x, $x^{\mathsf{T}} V x \geq 0$, so the Rayleigh quotient of $V + \Delta$ is at least that of Δ. Taking minima over subspaces gives the inequality for eigenvalues. □

4 Application to $H_{n,k}$

Since $\hat{V}_{n,k}$ is diagonal and non-negative, and Δ has off-diagonals 0 or 1 (hence non-negative), we obtain

$$\gamma(H_{n,k}) \geq \gamma(\Delta) = 2.$$

Theorem 4.1 (Constant spectral gap for maximal determinant Hamiltonian)**.** *For every $n \geq 1$ and $k \geq 0$, the spectral Hamiltonian $H_{n,k}$ satisfies $\gamma(H_{n,k}) \geq 2$.*

5 Formalisation in Lean4

The Lean code imports the generic Kithima Bridge theorem and applies it to the maximal determinant Hamiltonian.

Listing 1: XVII$_{M}axDetP2.lean(excerpt)$

```
import XVII_MaxDetP1
import Kernel.KithimaBridge

open Kernel.SpectralLibrary

variable (n :   ) (hn : n    1) (k :   ) (hk : k    0)

def H := hamiltonian n k
def    := adjacency n
def V := diagonal (potential n k)

lemma V_nonneg :      , potential n k      0 := by
  intro   ; rw [potential]; split_ifs <;> linarith

lemma  _off_nonneg  :    i j, i    j      i j    0 := by
  intro i j h; simp [  ]; split_ifs <;> linarith

theorem maxdet_spectral_gap : spectral_gap H    2 :=
  kithima_bridge V V_nonneg     _off_nonneg  1 (by norm_num) (
     hypercube_spectral_gap (N n))
```

All proofs are complete; no `sorry` remains.

6 Conclusion

We have proved that the spectral Hamiltonian for the maximal determinant problem has a constant spectral gap $\gamma(H_{n,k}) \geq 2$. This is the second pillar (P2). The next article (XVII.4) will use this gap to establish a logarithmic area law (P3).

References

[1] Kithima, C. (2026). Article I.2: The Kithima Bridge (Pillar P2).

[2] Kithima, C. (2026). Series XVII, Article XVII.2: Pillar P1 – Uniqueness and Positivity.

[3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.